

House Price Prediction using Machine Learning

Project Report

Submitted By

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Project Title

House Price Prediction using Machine Learning

Abstract

The House Price Prediction project is a Machine Learning based web application developed to predict the estimated price of a house using important features such as area and number of bedrooms. The system uses the Linear Regression algorithm to train the model and generate predictions.

The project integrates Machine Learning with Flask web development to create an interactive and user-friendly web application. Users can enter house details through the web interface and instantly receive the predicted house price.

This project demonstrates the practical implementation of Machine Learning, data processing, regression analysis, and web application development.

Introduction

House prices vary depending on many factors such as location, area, number of bedrooms, and facilities. Predicting accurate house prices manually can be difficult and time-consuming.

The purpose of this project is to automate the process of predicting house prices using Machine Learning techniques. The project uses a trained Linear Regression model to predict prices based on user inputs.

The project also provides a responsive web interface built using Flask, HTML, and CSS.

Objectives

The main objectives of this project are:

- To understand Machine Learning concepts
 - To implement Linear Regression algorithm
 - To create a predictive model for house prices
 - To integrate Machine Learning with Flask
 - To design a user-friendly web application
 - To deploy and manage the project using GitHub
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Problem Statement

Predicting house prices manually is complex because multiple factors influence the final price. Buyers and sellers need a system that can estimate house prices quickly and accurately.

This project solves the problem by using Machine Learning to analyze house features and predict estimated prices automatically.

Technologies Used

Technology	Purpose
Python	Core Programming
Flask	Backend Framework
scikit-learn	Machine Learning
Pandas	Data Processing

Technology	Purpose
NumPy	Numerical Operations
HTML/CSS	Frontend Design
Git & GitHub	Version Control

Machine Learning Algorithm

Linear Regression

Linear Regression is a supervised Machine Learning algorithm used to predict continuous numerical values.

The prediction formula used is:

$$\hat{y} = b_0 + b_1x_1 + b_2x_2$$

Where:

- x_1 = Area
- x_2 = Bedrooms
- $\hat{y}$ = Predicted Price
- b_0 = Intercept
- b_1, b_2 = Coefficients

The algorithm learns the relationship between input features and output values from the dataset.

Dataset Description

The dataset contains house details such as:

Area	Bedrooms	Price
1000	2	5000000
1500	3	7500000
2000	4	12000000

The dataset is stored in a CSV file named:

```
house_data.csv
```

System Architecture

```
User → Web Interface → Flask Backend → ML Model → Prediction Result
```

Project Structure

```
House-Price-Prediction/  
|  
├─ app.py  
├─ model.py  
├─ model.pkl  
├─ house_data.csv  
├─ requirements.txt  
|  
├─ templates/  
│   └─ index.html  
|  
├─ static/  
│   └─ style.css
```

Working of the Project

Step 1: Dataset Loading

The dataset is loaded using Pandas.

Step 2: Model Training

The Linear Regression model is trained using area and bedrooms as input features.

Step 3: Saving Model

The trained model is saved using Pickle.

Step 4: Flask Integration

The Flask application loads the trained model.

Step 5: User Input

The user enters area and bedroom values.

Step 6: Prediction

The model predicts the estimated house price.

Step 7: Result Display

The predicted price is displayed on the webpage.

Features of the Project

- Real-time house price prediction
 - Machine Learning integration
 - Flask web application
 - Responsive frontend design
 - Dark and Light theme support
 - Beginner-friendly implementation
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Advantages

- Fast predictions
 - User-friendly interface
 - Easy to understand
 - Lightweight application
 - Useful for learning Machine Learning concepts
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Limitations

- Uses small dataset
- Prediction depends on limited features
- Accuracy can improve with larger datasets
- Does not include real-time market data

Future Enhancements

Future improvements that can be added:

- Location-based prediction
 - Better UI/UX
 - Real-world datasets
 - Random Forest and XGBoost algorithms
 - Cloud deployment
 - Android application
 - Interactive graphs and charts
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Installation and Usage

Install Dependencies

```
pip install -r requirements.txt
```

Run the Application

```
python app.py
```

Open in Browser

```
http://127.0.0.1:5000
```

GitHub Repository

Project Repository:

<https://github.com/jatinrajvanshi94/House-Price-Prediction>

Learning Outcomes

Through this project, the following concepts were learned:

- Machine Learning fundamentals
 - Linear Regression
 - Flask framework
 - Frontend and backend integration
 - GitHub project management
 - Web application deployment basics
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Conclusion

The House Price Prediction project successfully demonstrates the implementation of Machine Learning in real-world applications. The project combines data analysis, Linear Regression, and Flask web development to create a functional prediction system.

This project helped in understanding how Machine Learning models are trained, integrated into web applications, and deployed for user interaction.

References

- Python Documentation
 - Flask Documentation
 - scikit-learn Documentation
 - Pandas Documentation
 - NumPy Documentation
 - GitHub Documentation
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Thank You